Junghyeon Kim (Andy)

iOS & XR Developer · Haptics Researcher

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I started as an iOS and XR app developer and became a haptics researcher, working on interfaces you can feel and not just look at. I build on-body haptic hardware and the software behind it, and I ship apps to the App Store.

Education

Kyung Hee University, M.S. in XR-AI Convergence 2025.03 — present
Haptics and VR Lab, advisor Prof. Seokhee Jeon

Apple Developer Academy @ POSTECH 2023.03 — 2023.12
Nine-month Challenge Based Learning program

Sungkyul University, B.S. in Computer Engineering 2017.03 — 2025.02

Seoul Gangseo High School 2014.03 — 2017.02

Publications & Presentations

Real-Time Body-Part Classification from a Single IMU for Adaptive On-Body Haptic Feedback
Junghyeon Kim, M. S. Hashem, M. I. Awan, S. Jeon
AsiaHaptics 2026 (Manuscript 59) · first author · Accepted, 2026.11

Soft Pneumatic Actuator for AR, VR, and Feel-Through Sensation
M. S. Hashem, A. Raza, **Junghyeon Kim**, S. Jeon
AsiaHaptics 2026 (Manuscript 32) · co-author · Accepted, 2026.11

Push-Latch-Inspired Vibrotactile Feedback for Handleless Car Doors: Markov Analysis of First-Time Interaction
T. Ogay, A. Abdulali, M. S. Hashem, **Junghyeon Kim**, L. Cai, S. Jeon
AsiaHaptics 2026 (Manuscript 31) · co-author · Accepted, 2026.11

IMU-Based Real-Time Body-Part Classification for Adaptive On-Body Haptic Feedback
Junghyeon Kim, M. S. Hashem, M. I. Awan, S. Jeon
4th Korea Haptics Conference, Session O1, Seoul · first author, oral presentation · 2026.08

Effects of K-POP Musical Element-Based Audio-to-Haptic Feedback on Haptic Experience in K-POP Virtual Reality Concerts
G.-R. Kim, H. Jang, **Jung-Hyeon Kim**, T. Woo
Journal of the Korea Entertainment Industry Association (JKEIA), Vol. 19, No. 1, pp. 1–12 · co-author · Published, 2026.02

Patent

Method and apparatus for providing escape-room content in a CAVE-based multi-projection XR environment
Filed 2026.02
Korean Patent Application No. 10-2026-0024399 · co-inventor · Kyung Hee University Industry-Academic Cooperation Foundation

Awards

3rd Place, Student Innovation Challenge, IEEE World Haptics Conference 2025 2025.06
“Haptic Latch: Enhancing Passenger Safety with Haptic Feedback in Vehicle Door Mechanisms”

Grand Prize, In-house (JAVA) Competition 2020.10
Health Keeper

Released Apps & Projects

IOOI · App Store 2026.10 — 2026.11

A decalcomania app for the foldable iPhone Duo. The inner display is the paper you paint on, the outer display is its back, and only the spots you press print in mirror image when you unfold. Includes an iMessage sticker extension. SwiftUI, MVVM, Apple Pencil.

Touchstone · App Store · Web 2026.09 — present

An iPhone that works as a black trackpad for a Mac and buzzes when the cursor rests on false information in a web

page. iPhone app, Mac app, Chrome and Safari extension, verification engine, and 3D-printable holders. SwiftUI, Core Haptics, TypeScript.

Feel the Place, WWDC 2026 Swift Student Challenge 2026.02

A single-screen app that lets you feel a place through your fingertip. SwiftUI, Core ML, Vision, Core Haptics.

Surface Haptic 2026.01

A Safari Web Extension that gives images on a web page a surface you can feel. Swift, AppKit, JavaScript.

AR Knob 2025.12

A research tool for simulating physical controls purely with haptic feedback. SwiftUI, Core Haptics, Core Bluetooth, Core Motion.

Ilsang · **App Store** 2024.06 — present

A team app live on the App Store that offers quests to turn repetitive daily routines into something fresh. iOS, SwiftUI, AWS, Alamofire, Swift Concurrency.

PostKit · **App Store** 2023.09 — present

A team app live on the App Store that helps small café owners who want to promote on social media but lack ideas. iOS, SwiftUI, CoreData.

Earlier team projects 2020 — 2023

Dots (2023), MatQ (2023), DiscoverMe (2023), who?Yak (2021–2022), DumChoeMyeot? (2021), Health Keeper (2020).

Skills

- **Haptics & hardware:** Core Haptics, ERM/LRA, SMA and pneumatic actuators, silicone casting, FSR/IMU/proximity sensing, Arduino Uno, Teensy 4.1, DAQ, 3D printing
- **Apps:** Swift, SwiftUI, UIKit, Swift Concurrency, Core ML and Vision, ARKit, RealityKit, visionOS, Unity, Kotlin and Java
- **Data & web:** Python (PyTorch, Core ML Tools), C/C++ for embedded firmware, TypeScript, JavaScript, HTML/CSS

Volunteer Experience

Coding Mentor, SW Creative Camp Student Volunteer Corps, CJ UNIT 6th (CJ OliveNetworks) 2022

Coding Mentor, “Be a Coding Hero”, Microsoft 2017 & 2018

Museum Volunteer, “Google Museum on the Go”, Google & National Museum of Korea 2017

Other

Social Service Personnel, Seoul Metro 2018.08 — 2020.06

Languages: Korean (native), English (professional working proficiency), Japanese (limited working proficiency)